

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
1	(a)	(i)	$\rho = \frac{45.4}{5.6} = 8.1 \text{ [g cm}^{-3}\text{]}$ (1) Accept 8.11 but not 8.10 or 8.12) $p_{\text{volume}} = 3.6\%$ and $p_{\text{mass}} = 1.1\%$ or $p_{\text{density}} = \frac{0.2}{5.6} + \frac{0.5}{45.4}$ or $= \frac{0.2}{5.6} \times 100 + \frac{0.5}{45.4} \times 100$ (1) $p_{\text{density}} = 4.7\%$ (ecf : p_{volume} and p_{mass})(Adding % uncertainties) (1) No sig fig penalty in % unc e.g. allow 4.67% Alternative for 2nd and 3rd marks $\text{Unc} = \frac{\text{max} - \text{min}}{2} = \frac{8.50 - 7.74}{2}$ (1 mark for either 8.50 or 7.74 or both) Correct method to calculate % unc e.g. $\frac{7.74}{8.11} \times 100 = 95.4\% \sim 4.6\%$ (1)	1	1		3	2	3
		(ii)	0.38 if 4.7% used or 0.41 if 5% used. Accept 0.4[0] Allow 1 or 2 sig figs ecf on sig figs from (i) Bod on incorrect % unc from (i)		1		1	1	1
	(b)	(i)	Iron and Brass and Nichrome (all required) (1) ecf from (a)(ii) All lie within calculated uncertainty(1) Accept in the range 7.7 to 8.5 Do not accept large uncertainty only or vague reference to uncertainty		1	1	2		2
		(ii)	Volume (1) Greater % uncertainty or linked to 3.6% (1) Don't accept reference to absolute uncertainties			2	2		2
			Question 1 total	1	4	3	8	3	8